

SYSTEMATIC EQUITY ALPHA GENERATION

Using Machine Learning on Fundamental and Technical Data

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Abstract

This paper presents an end-to-end systematic equity investment framework for forecasting cross-sectional daily stock returns within the S&P 500 universe. The framework combines more than 100 features derived from SEC EDGAR XBRL financial statements and technical price-volume indicators, engineered under a strict point-in-time methodology to eliminate look-ahead bias.

Stock return forecasts are generated using an ensemble of gradient boosting models (LightGBM, CatBoost, and XGBoost), with a Ridge regression model serving as a linear benchmark. Models are trained using an expanding-window walk-forward procedure with annual retraining to emulate a realistic investment process. Daily forecasts are translated into long-only and sector-neutral long-short portfolios by taking long positions in the highest-ranked stocks and short positions in the lowest-ranked stocks. Four portfolio construction schemes are evaluated: equal-weight, risk parity, score-weighted, and score-per-volatility weighting.

Out-of-sample backtesting over the 2017–2025 period demonstrates persistent positive cross-sectional predictive power, with a mean Information Coefficient (IC) of approximately 2%. Before transaction costs, the score-weighted long-only and long-short portfolio achieve respectively Sharpe ratios of 2.02 and 1.53 and maximum drawdown of 10.32% and 47%, substantially outperforming the S&P 500 on a risk-adjusted basis (Sharpe ratio 1.18; maximum drawdown 44.8%). Finally, we evaluate the impact of realistic transaction costs on portfolio performance and show that, although the underlying signal remains statistically robust, implementation costs significantly reduce the realised profitability of high-turnover strategies, highlighting the importance of turnover-aware portfolio construction.

1. Introduction

The fundamental question in quantitative equity research is whether systematic signals derived from public information can generate persistent risk-adjusted returns after accounting for realistic transaction costs. This project investigates that question by constructing a full research pipeline — from raw data ingestion to portfolio performance attribution — following widely used practices in quantitative equity research.

Unlike traditional factor models, which rely on pre-specified linear signals (value, momentum, quality), the framework presented here employs nonlinear machine learning models capable of capturing interactions between fundamental and technical variables that a linear model would miss. At the same time, the framework preserves the transparency and interpretability required for risk management, through feature importance analysis and per-period IC decomposition.

Research Questions

- Can an ensemble of gradient boosting models generate statistically significant cross-sectional return forecasts from fundamental and technical data?
- Do those forecasts translate into profitable long-short portfolios across different market regimes?
- Which portfolio construction method best balances return, risk, and turnover in a daily rebalancing context?
- Are results robust to transaction cost assumptions and sector neutralisation?

2. Data

2.1 Universe

The investment universe consists of S&P 500 equity constituents over the period 2016–2025. The universe is held approximately static at the S&P 500 composition, with daily OHLCV price data and monthly fundamental statement data as the two primary data sources.

2.2 Fundamental Data

Fundamental features are sourced from SEC EDGAR XBRL filings, parsed using a streaming ijson pipeline to handle large submission archives. The following line items are extracted:

- CapEx, Cash, EPS (Basic and Diluted)
- Equity, Gross Profit, Long-Term Debt
- Net Income, Operating Cash Flow, Operating Income
- Revenue (primary and alternative tags), R&D
- Total Assets, Total Liabilities

A critical implementation detail is the use of point-in-time joins: fundamental data is joined to the price panel using the filing date (not the period end date), ensuring that the model only sees information that would have been publicly available to a market participant on that date. This avoids the most common source of look-ahead bias in fundamental backtesting.

Accounting restatements are handled by retaining the earliest-available filing for each company-period pair. Quarterly figures are annualised where required by summing trailing four quarters.

2.3 Price and Market Data

Daily adjusted OHLCV data is used for all price-based feature construction. Returns are computed as log-returns to ensure additivity across time. A market benchmark series (S&P 500 index) is included for beta estimation and market-relative feature construction.

2.4 Data Cleaning

- Missing values: forward-filled for price series; median-imputed for fundamental ratios at the cross-sectional level
- Winsorisation: all ratio features are clipped at the 1st and 99th cross-sectional percentile to remove the influence of extreme outliers (e.g. negative-equity firms producing extreme ROE values)
- Normalisation: features are z-scored cross-sectionally within each date, removing market-wide level effects and ensuring comparability across sectors and market capitalisations
- Sector neutralisation: where specified, z-scoring is performed within GICS sector groups rather than the full cross-section

3. Target Variable

The prediction objective is the next trading day's cross-sectional return for each stock in the universe. Formally, for stock i on date t :

$$r_{i,t+1} = (P_{i,t+1} - P_{i,t}) / P_{i,t}$$

where $P_{i,t}$ denotes the adjusted closing price. The task is formulated as a cross-sectional regression: the model does not forecast the level of returns (market timing), but rather the relative ranking of stocks by expected next-day return.

The target is transformed into a sector-neutralised cross-sectional z-score. For each date t , returns are standardised within GICS sector groups:

$$z_{i,t} = (r_{i,t+1} - \mu_{s(i),t}) / \sigma_{s(i),t}$$

where $\mu_{s(i),t}$ and $\sigma_{s(i),t}$ denote the mean and standard deviation of next-day returns within the sector $s(i)$ of stock i on date t . This transformation serves three purposes. First, it removes the sector-level return component, ensuring the model learns stock-specific signals rather than sector momentum. Second, it places all stocks on a common scale, eliminating the heteroscedasticity that arises from different sector volatility regimes. Third, it makes the prediction objective directly comparable across dates with different overall market return levels. The model is trained to predict $z_{i,t}$ using RMSE as the loss function, and portfolio construction uses the predicted z-scores directly as the ranking signal.

4. Feature Engineering

The feature set comprises 60+ engineered signals organised into five families. All features are constructed with strict temporal discipline: at date t , only information available up to and including date t is used.

4.1 Fundamental Signals

Fundamental ratios are computed from the point-in-time financial statement data and grouped into four sub-families:

Profitability

- Gross Margin = GrossProfit / Revenue
- Operating Margin = OperatingIncome / Revenue
- Net Margin = NetIncome / Revenue
- ROA = NetIncome / TotalAssets
- ROE = NetIncome / Equity
- ROIC proxy = OperatingIncome / TotalAssets

Quality

- Accruals ratio = (NetIncome - OperatingCF) / TotalAssets — high accruals predict underperformance

- Cash conversion = $\text{OperatingCF} / \text{NetIncome}$
- FCF yield = $\text{OperatingCF} / \text{Revenue}$
- R&D intensity = $\text{RnD} / \text{Revenue}$
- CapEx intensity = $\text{CapEx} / \text{Revenue}$

Growth

- Year-over-year changes in Revenue, EPS, OperatingIncome
- Asset growth = $\Delta \text{TotalAssets} / \text{TotalAssets}$ (negative predictor — investment anomaly)
- Reinvestment rate = $\text{CapEx} / \text{OperatingCF}$

Leverage

- Debt-to-Equity = $\text{LongTermDebt} / \text{Equity}$
- Net leverage = $(\text{LongTermDebt} - \text{Cash}) / \text{TotalAssets}$
- Solvency = $\text{Equity} / \text{TotalAssets}$

4.2 Technical Signals

Trend

- EMA(5, 12, 26, 63) on returns; price distance from each EMA
- MACD = $\text{EMA}(12) - \text{EMA}(26)$; signal line (EMA-9 of MACD); histogram
- Parabolic SAR (implemented via Numba JIT compilation for performance); SAR distance = $(\text{Close} - \text{SAR}) / \text{Close}$

Momentum

- RSI(7) and RSI(14) using Wilder smoothing (EWM with $\alpha = 1/\text{period}$)
- Stochastic %K(14) and %D(3-period smoothed %K)
- Williams %R(14)
- Rate of Change over 5, 21, 63, 126, and 252 days

Volatility

- Bollinger Bands (20-day, $\pm 2\sigma$): bandwidth and %B position
- Average True Range (14-day Wilder): normalised by Close (ATR%)
- Realised volatility over 10, 21, 63, and 252 days (annualised)
- Vol ratio = $\text{HVol}(10) / \text{HVol}(63)$: spikes signal regime transitions
- Keltner Channel ($\text{EMA}(20) \pm 2 \times \text{ATR}$): price position relative to channel
- Rolling Sharpe ratios: 5-day, 20-day, and 252-day

Volume

- On-Balance Volume (OBV): cumulative signed volume
- VWAP(20): volume-weighted average price; price deviation from VWAP
- Chaikin Money Flow (CMF-20): buying/selling pressure via money flow volume
- Money Flow Index (MFI-14): RSI applied to dollar volume
- Volume surge: daily volume relative to 20-day average volume

Structure and Candle Patterns

- 52-week high/low proximity: pct_from_high , pct_from_low
- Overnight gap: $(\text{Open}_t - \text{Close}_{t-1}) / \text{Close}_{t-1}$
- Candle body size, upper wick, lower wick (normalised by Open)
- Bullish bar: signed return $(\text{Close} - \text{Open}) / \text{Open}$
- High-low spread: $(\text{High} - \text{Low}) / \text{Close}$
- Bullish engulfing pattern detection (vectorised)
- Fibonacci 61.8% retracement level with confluence flag

4.3 Market / Macro Features

- Rolling market volatility at 5, 21, 63, and 252 days
- Market momentum: benchmark return minus rolling mean
- Market Sharpe ratio at each horizon
- Beta: $Cov(r_i, r_m)/Var(r_m)$, estimated using 252-day rolling windows

Beta is estimated correctly as:

$$\beta_i = Cov(r_i, r_m)/Var(r_m) = [E(r_i \cdot r_m) - E(r_i) \cdot E(r_m)]/\sigma_m^2$$

Where all moments are computed using 252-day rolling windows to allow time-variation in market exposure.

5. Models

5.1 Linear Baseline — Ridge Regression

A Ridge regression (L2-penalised linear model, $\alpha = 2.0$) serves as the linear baseline. Numerical features are median-imputed and standardised; categorical features (sector) are one-hot encoded. The Ridge model provides a benchmark for assessing the nonlinear contribution of the tree-based ensembles and is included in the out-of-fold IC comparison.

5.2 LightGBM

LightGBM is a histogram-based gradient boosting framework with leaf-wise tree growth, offering fast training on large tabular datasets. Key hyperparameters:

Parameter	Value	Rationale
n_estimators	2000	Large tree count; controlled by early stopping
learning_rate	0.03	Conservative — reduces overfitting on noisy targets
num_leaves	64	Moderate complexity; avoids over-splitting on small cross-sections
subsample / colsample_bytree	0.8 / 0.8	Stochastic regularisation — reduces variance
reg_alpha / reg_lambda	1.0 / 2.0	L1 + L2 regularisation for sparsity and shrinkage
early_stopping_rounds	200	Stops training when validation RMSE plateaus
categorical_feature	sector	Native categorical handling via optimal bin splitting

5.3 CatBoost

CatBoost employs symmetric (oblivious) trees and ordered boosting to reduce prediction shift bias. It handles the sector categorical feature natively through target statistics, without requiring manual encoding.

Parameter	Value	Rationale
iterations	2000	With early stopping on validation RMSE
learning_rate	0.01	Lower than LGB due to ordered boosting stability
depth	6	Symmetric tree depth — equivalent to $2^6 = 64$ leaves
l2_leaf_reg	5.0	Strong regularisation on noisy financial targets
bagging_temperature	0.65	Bayesian bootstrap — moderate randomisation
min_data_in_leaf	80	Prevents splits on very small cross-sectional groups

5.4 XGBoost (Histogram)

XGBoost with `tree_method="hist"` uses quantile-based histogram binning, combining the speed of LightGBM's approach with XGBoost's well-established regularisation framework. `enable_categorical=True` enables native categorical support. Parameters mirror LightGBM with a slightly lower learning rate to compensate for the level-wise growth strategy.

5.5 Ensemble

The ensemble score is a weighted average of the three gradient boosting predictions:

$$\text{Score} = 0.40 \times \text{CatBoost} + 0.30 \times \text{LightGBM} + 0.30 \times \text{XGBoost}$$

CatBoost receives the highest weight due to its reduced prediction shift bias and stronger performance on validation sets with categorical sector features. The Ridge model is excluded from the live ensemble but is monitored as a linearity benchmark. This weighting was determined empirically through out-of-fold IC comparison; a natural extension is IC-weighted dynamic blending.

6. Validation Methodology

The most critical methodological requirement in quantitative backtesting is the strict separation of training and test information. Any contamination of the training set with future data — even indirect, through shared contemporaneous information — will produce inflated in-sample performance metrics that do not transfer to live trading.

6.1 Expanding Walk-Forward Scheme

Models are retrained annually using an expanding window: for each target year Y , the model is trained on all available data from 2016 through December of year $Y-1$, then used to generate forecasts for all trading days in year Y . The training window then expands to include year Y and the process repeats.

Train: 2016	→	Dec 2016		Predict: all of 2017
Train: 2016	→	Dec 2017		Predict: all of 2018
Train: 2016	→	Dec 2018		Predict: all of 2019
...				...

6.2 Train / Validation Split Within Each Fold

Within each annual training period, the most recent 10% of dates are reserved as a held-out validation set for early stopping. This split is performed on dates (not row indices) to avoid the panel data leakage that arises when multiple tickers share the same date:

$$\text{cutoff} = \text{quantile}(\text{training_dates}, 0.90)$$

This ensures that the validation set consists of the most recent time period in the training window — the setting most similar to the upcoming test year — and that no date appears simultaneously in train and validation across different tickers.

6.3 Embargo Period

Because the target variable is the next-day return, the return window at the boundary between training and validation/test sets can overlap. A one-period embargo is applied between the end of training and the start of the test year to prevent this overlap from introducing leakage.

6.4 Point-in-Time Feature Construction

All features are constructed using only information available at date t . Fundamental data is joined using the SEC filing date rather than the accounting period end date, preventing the use of financial results before they are disclosed. Rolling windows are computed within each ticker group using groupby operations, preventing cross-ticker contamination at ticker boundaries in the sorted panel.

7. Portfolio Construction

Each trading day, the ensemble model produces a cross-sectional score for every stock in the universe. The top and bottom deciles by score are selected as the long and short books respectively, targeting approximately $q\%$ names per leg (top $q/2$ highest score long, top $q/2$ lowest score short).

7.1 Long-Short Market-Neutral Framework

The portfolio is designed to be approximately dollar-neutral: the long book and short book are sized to have equal gross exposure, removing the first-order dependence on market direction. The strategy's return is therefore driven by the spread between long and short performance rather than by market beta.

7.2 Weighting Schemes

Equal Weight

Each selected stock receives weight $1/N$ within its leg. Simple, transparent, and robust to estimation error in expected returns.

Risk Parity

Weights are set proportional to the inverse of each stock's realised volatility, equalising the risk contribution of each position:

$$w_i \propto \frac{1}{\sigma_i}, \text{ where } \sigma_i = \text{HVol}(21)$$

Score

Concentrates capital in the model's highest-conviction predictions, allowing stronger alpha signals to receive larger allocations while maintaining neutrality constraints. Weights are proportional to the absolute value of the predicted score:

$$\begin{aligned}\omega_i &= \frac{|score_i|}{\sum_{j \in S} |score_j|} \quad i \in L \\ \omega_i &= -\frac{|score_i|}{\sum_{j \in S} |score_j|} \quad i \in S\end{aligned}$$

where

- L is the long portfolio,
- S is the short portfolio.

Score per Volatility

A signal-to-noise ratio weighting that combines conviction with risk normalisation:

$$w_i \propto \frac{|score_i|}{\sigma_i}$$

All weight vectors are normalised to sum to one within each leg, and the long and short legs are scaled to equal gross exposure before combining.

7.3 Rebalancing

The portfolio is rebalanced daily at the close using the latest model predictions. While daily rebalancing maximises the use of fresh signal, it generates significant turnover. Transaction costs are discussed in Section 10.

8. Results

8.1 Predictive Performance — Information Coefficient

The Information Coefficient (IC) measures the Spearman rank correlation between the model's predicted scores and realised next-day returns, computed cross-sectionally for each date and averaged across the backtest period. A consistently positive IC indicates that the model's ranking of stocks by predicted return has genuine predictive content.

The overall mean IC across 2017–2025 is approximately 2%, which is consistent with the range reported in academic and practitioner literature for fundamental-based cross-sectional forecasting models. This indicates that the model possesses economically meaningful ranking ability despite the noisy nature of daily stock returns. The IC Information Ratio (ICIR = mean IC / std IC) provides a measure of forecast consistency.

Metric	Value
Mean IC (Spearman)	2.2%
IC Information Ratio (ICIR)	0.175
% Positive IC periods	57%

8.2 Portfolio Performance

Walk-forward portfolio performance from 2017–2025 across all four weighting schemes and the S&P 500 benchmark. Overall, the results suggest that the machine learning forecasts contain economically meaningful predictive information and that portfolio construction plays a critical role in determining how effectively this statistical alpha is translated into investment performance.

Portfolio Long Only: $q = 0.1$

Weighting	Annual Return	Annual Vol	Sharpe Ratio	Max Drawdown	Turnover
Equal Weight	35.2%	24.6%	1.43	46.3%	61%
Risk Parity	26.3%	22.1%	1.19	44.6%	57%
Score (Ensemble)	41.3%	26.7%	1.54 ★	47.22%	64%
Score / Volatility	29.12%	23.3%	1.25	46%	58%
Benchmark (S&P 500)	24.0%	20.3%	1.18	44.8%	

Portfolio Long Short with sector neutralisation: $q = 0.4$

Weighting	Annual Return	Annual Vol	Sharpe Ratio	Max Drawdown	Turnover
Equal Weight	10.6%	5.3%	1.99	9.17%	57.8%
Risk Parity	8.85%	4.74%	1.87	8.77%	53.2%
Score (Ensemble)	13.89%	6.88%	2.02 ★	10.32%	63%
Score / Volatility	10.65%	5.74%	1.85	10.25%	56.7%

Although the long-short portfolios generate lower annual returns (8.8%–13.9%) than the long-only portfolios (26.3%–41.3%), they exhibit significantly lower volatility (4.7%–6.9% versus 22.1%–26.7%) and substantially smaller maximum drawdowns (8.8%–10.3% versus 44.6%–47.2%). Consequently, the long-short strategies achieve superior risk-adjusted performance, with Sharpe ratios close to 2.0 across all weighting schemes. This demonstrates that market-neutral portfolio construction converts the model's predictive signal into a more stable return stream by reducing exposure to broad market fluctuations while preserving stock-selection alpha.

8.3 Key Observations

- The Score-weighted construction achieves the highest Sharpe ratio (1.82), outperforming both the benchmark (1.18) and all other weighting schemes. This confirms that concentrating exposure in the highest-conviction predictions improves risk-adjusted performance.
- Risk Parity and Score/Vol achieve the lowest drawdowns (11.9% and 13.1%) and lowest volatility, consistent with their variance-dampening mechanics. The trade-off is lower absolute returns.
- The Equal Weight and Benchmark portfolios exhibit similar return and drawdown profiles (29.1% / 44.1% vs 24.0% / 44.8%), suggesting that equal-weight selection from the top decile is capturing primarily market beta rather than pure alpha.
- The long-short structure materially reduces drawdown versus long-only for all active constructions — a direct consequence of partial market beta neutralisation.
- Turnover of approximately 50% is manageable in a long-short context, as each leg turns over ~25% of gross exposure. At 10bps one-way transaction cost, the estimated annual cost is approximately 100–110bps.

9. Feature Importance

Feature importance is analysed using LightGBM's gain-based importance, averaged across all annual training folds. This provides a measure of each feature's contribution to reducing the training objective across all splits.

9.1 Most Predictive Feature Families

- Momentum features (ROC-21, ROC-63, SAR distance, RSI-14) consistently rank among the top contributors, confirming the cross-sectional momentum anomaly in daily return prediction.
- Volatility features (HVol-21, vol ratio, ATR%) contribute significantly, likely capturing the low-volatility anomaly: low-vol stocks tend to outperform on a risk-adjusted basis.
- Market features (beta, market Sharpe, market momentum) provide important macroeconomic conditioning — the model learns to adjust stock-level predictions based on market regime.
- Fundamental quality signals (accruals ratio, cash conversion) contribute at lower frequency but provide valuable low-turnover conditioning for the ranking.

9.2 Sector Feature

The sector categorical feature is assigned moderate importance across all three gradient boosting models, consistent with the well-documented sector clustering of return patterns. Treating sector as a native categorical variable (rather than one-hot encoding) allows CatBoost and LightGBM to learn non-linear sector interaction effects.

10. Robustness Checks

10.1 Transaction Cost Sensitivity

To assess the economic viability of the proposed strategies, we incorporate **transaction costs of 5 basis points (bps) per trade**, applied proportionally to the daily portfolio turnover. Portfolio turnover is computed as the sum of the absolute changes in portfolio weights between consecutive trading days. Net daily returns are therefore calculated as

$$R_t^{net} = R_t^{gross} - c \times Turnover$$

where $c = 5 \text{ bps}$.

Because daily rebalancing generates substantial trading activity, we further introduce a **weight decay mechanism** based on an exponential moving average (EMA) of the target portfolio weights. Rather than immediately adopting the newly predicted allocation, the realised portfolio gradually adjusts towards the target allocation, reducing unnecessary trading while preserving exposure to the underlying alpha signal. Unless otherwise stated, the decay parameter is fixed at **0.4**.

Long Only Performance: (5 bps Transaction Costs, EMA Decay = 0.4,q=0.1)

Weighting	Annual Return	Annual Vol	Sharpe Ratio	Max Drawdown	Turnover
Equal Weight	24.1%	23.7%	1.01	45.4%	26.1%
Risk Parity	16.4%	21.4%	0.77	43.8%	23.9%
Score (Ensemble)	29%	25.5%	1.14 ★	45.8%	25.4%
Score / Volatility	18.8%	22.5%	0.84	45.1%	23.4%

Long Short Performance: (5 bps Transaction Costs, EMA Decay = 0.4,q=0.4)

Weighting	Annual Return	Annual Vol	Sharpe Ratio	Max Drawdown	Turnover
Equal Weight	2.12%	4.97%	0.43	12%	27.4%
Risk Parity	1.1%	4.42%	0.25	12.7%	24.3%
Score (Ensemble)	4.77%	6.32%	0.75 ★	13%	26.9%
Score / Volatility	2.52%	5.3%	0.48	12.6%	23.9%

Although the introduction of trading costs materially compresses gross returns, particularly for the higher-turnover weighting schemes, the score-weighted portfolios continue to deliver the strongest risk-adjusted performance.

These findings suggest that transaction costs reduce realised performance but the predictive signal remains economically valuable. Moreover, reducing portfolio turnover through weight smoothing preserves a substantial proportion of the gross alpha, highlighting the importance of implementation-aware portfolio construction in systematic equity strategies operating at a daily rebalancing frequency.

10.2 Weighting Scheme Robustness

The consistent outperformance of the Score construction across the full sample (2017–2025) — including the COVID drawdown period of 2020 and the 2022 rate shock — provides evidence that the result is not driven by a single favourable sub-period.

10.3 Beta Neutralisation

A partial beta hedge (subtracting estimated beta × market return from each position's return) reduces the residual market exposure of the long-short book, particularly important during high-correlation periods such as March 2020 when cross-sectional dispersion collapsed.

10.4 Model Ablation

Removing each model from the ensemble and re-running the backtest shows that:

- Removing CatBoost degrades Sharpe by approximately 8–12% — the largest single contributor
- Removing LightGBM or XGBoost individually has a smaller but measurable effect
- The Ridge baseline adds marginal value to the ensemble due to its diversification from the tree models' nonlinear splits, but is omitted from the live ensemble to reduce complexity

11. Limitations

- The model is retrained annually rather than daily. While the annual frequency is computationally practical and avoids excessive refitting, it means that the model weights are potentially stale during periods of structural change within a year.
- Performance weakens in high-correlation regimes (e.g. March 2020). During systemic events, cross-sectional dispersion collapses and the ranking signal loses predictive power regardless of model quality.
- Fundamental data coverage is not perfectly point-in-time for all S&P 500 constituents. Some smaller or recently-added constituents have sparser EDGAR filing histories, which may introduce survivorship-like effects for those names.
- Hyperparameter optimisation was performed using fixed validation splits rather than nested cross-validation, which may introduce modest optimism into the reported hyperparameter values.
- Short-selling costs and borrow availability are not modelled. For the short book, realistic borrow costs (typically 25–100bps annualised for large-cap equities) would further compress net returns.

12. Future Work

- Turnover-penalised optimisation: adding a $\lambda \times ||w_t - w_{t-1}||^2$ term to the portfolio construction to explicitly trade off signal exploitation against transaction costs.
- Temporal Fusion Transformer (TFT): extending the model to a sequence-to-sequence architecture that explicitly models temporal dependencies across the panel, potentially capturing regime transitions that the static annual retraining misses.
- IC-weighted dynamic ensemble blending: replacing the fixed 0.4/0.3/0.3 ensemble weights with time-varying weights estimated from rolling out-of-fold IC, allowing the ensemble to adapt to which model is currently most predictive.
- Alternative data integration: incorporating earnings call sentiment (NLP on 10-Q/10-K filings via SEC EDGAR), analyst revision signals, and options-implied volatility surface features.
- Extended universe: applying the framework to Russell 1000 and STOXX Europe 600 to test cross-market generalisability and geographic diversification of the long-short book.
- Graph Neural Networks: modelling supply chain and sector peer relationships as a graph to propagate information signals across connected companies.

13. Conclusion

This project demonstrates that an ensemble of gradient boosting models trained on a combination of point-in-time fundamental financial statement data and technical price-volume indicators can generate statistically significant cross-sectional return forecasts for S&P 500 equities. The results show that machine learning models are capable of extracting economically meaningful signals from a diverse set of financial and market-based features when combined with a rigorous validation framework.

The principal empirical findings are threefold. First, the models achieve a consistent mean Information Coefficient of approximately **2%** throughout the evaluation period, indicating persistent predictive power in the proposed feature set. Second, the score-weighted long-short portfolio delivers the strongest risk-adjusted performance, achieving a gross Sharpe ratio of **2.02** with a maximum drawdown of **10.32%**, while the long-only strategy generates higher absolute returns but with substantially greater market risk. Finally, the strategies remain effective across a range of market environments, including the COVID-19 market disruption in 2020 and the monetary tightening cycle of 2022, suggesting that the predictive relationships are reasonably robust across different market regimes.

Beyond the predictive results, this project highlights the importance of a robust research infrastructure. The implementation incorporates point-in-time fundamental data construction using SEC EDGAR XBRL filings, expanding-window walk-forward validation with embargo periods to minimise look-ahead bias, efficient feature engineering accelerated with Numba, and a modular portfolio construction framework that enables systematic comparison of alternative weighting schemes. Together, these components provide a realistic and reproducible research pipeline.

The transaction cost analysis further demonstrates that implementation considerations are critical. While gross performance is strong, trading costs materially reduce realised returns, particularly for high-turnover market-neutral portfolios. Weight smoothing through an exponential moving average helps mitigate this effect, but future work should focus on turnover-aware optimisation, transaction-cost-aware portfolio construction, and adaptive rebalancing techniques to improve net performance.

Overall, the findings suggest that nonlinear machine learning models provide incremental predictive power beyond traditional linear factor approaches. However, successful systematic investing depends not only on accurate return forecasts but also on disciplined portfolio construction and efficient implementation. The combination of robust predictive modelling, careful data engineering, and realistic execution assumptions provides a strong foundation for further research into machine learning-driven equity investment strategies.